



Precision Low Power: Dry Electrode Challenges in Biopotential Signal Chains



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4 minute read time.

In the [last blog post](#), I talked about the difference between two and three electrode solutions when biasing the measurement inputs and managing CMRR performance. Now I'd like to focus a bit more on the electrodes themselves. One question I have heard a few times with respect to making biopotential measurements is "What is the sensor?". I discussed a bit about how biopotential signals are generated in my [first blog](#) and that they can be measured at the skin with electrodes. These electrodes act as transducers for ionic currents in the body and can be as simple as placing a conductive material such as metal on the skin.

In fact, the first "electrodes" I ever used were two pairs of metal scissors! It was a little over ten years ago and I was testing a first prototype of the [AD8232](#) which was the precursor to the [AD8233](#) discussed in the [third blog](#). What better way to check the signal conditioning performance, than to see your own ECG signal on the oscilloscope (isolated and battery powered for safety of course). I attached two wires to the evaluation board, and then shorted those wires to the scissors. This made it easier to make good contact when measuring ECG at the hands. Eventually, I upgraded to a set of electrodes used in exercise equipment (see Figure 1). You can see that in this case, there are 4 electrodes (2 per hand). The top pieces of metal are there for making the differential biopotential

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measurement. The other two on the bottom are used for biasing the body and/or right leg drive, demonstrating you can do this at multiple points. The wires coming back from the electrodes are shielded to minimize interference.

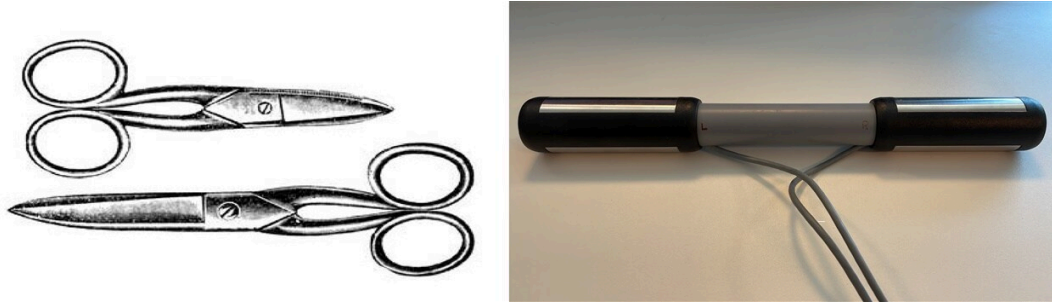


Figure 1 – Scissors “electrodes” and exercise equipment electrodes

Electrode Model

So how can we electrically model these electrodes? Figure 2 shows an example model where “ E_{hc} ” is the material dependent half-cell potential which is the result of dissimilar electrolytic interfaces. This is in series with an impedance (R_d in parallel with C_d) that represents the electrode-skin interface and polarization at this location. This is also in series with another resistor (R_s) that takes into account other factors such as the resistance of the electrode material.

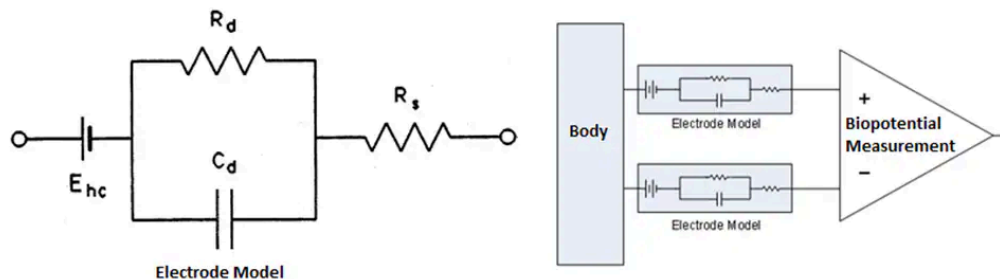


Figure 2 – Biopotential electrode model and location in measurement solution [1]

When referring to polarization, a perfectly “non-polarizable” electrode behaves more like the resistor “ R_d ”. A close example of this is the silver-silver chloride (Ag/AgCl) electrodes typically used in medical applications where an electrolytic gel may be applied. These are better for measuring, since they have lower contact impedance, lower noise, and reduced motion artifacts. Ag/AgCl electrodes also have a lower half-cell potential relative to other materials as shown in Figure 3 below.

Perfectly “polarizable” electrodes behave more like the capacitor “ C_d ”. A close example of this would be Platinum. These electrodes are better for stimulation and typically have higher noise and worse motion artifacts. In reality it is difficult to manufacture a purely polarizable or non-polarizable electrode so there will always be some R_d in parallel with C_d .

One clarification on half-cell potentials is that this is not the DC offset you are measuring differentially at the electrodes. Each electrode has a half-cell potential, so what you are measuring is the mismatch or difference between these. For perfectly matched electrodes, the half-cell potential is part of the common mode voltage. Also, this half-cell potential changes as current flows through the electrode due to polarization. This amount of change is called the overpotential.

Half-cell Potentials for Materials and Reactions Encountered in Biopotential Measurement	
Metal and Reaction	Half-cell Potential, V
$\text{Al} \rightarrow \text{Al}^{3+} + 3\text{e}^{-}$	-1.706
$\text{Ni} \rightarrow \text{Ni}^{2+} + 2\text{e}^{-}$	-0.230
$\text{H}_2 \rightarrow 2\text{H}^{+} + 2\text{e}^{-}$	0.000 (by definition)
$\text{Ag} + \text{Cl}^{-} \rightarrow \text{AgCl} + \text{e}^{-}$	+0.223
$\text{Ag} \rightarrow \text{Ag}^{+} + \text{e}^{-}$	+0.799
$\text{Au} \rightarrow \text{Au}^{+} + \text{e}^{-}$	+1.680

Figure 3 – Table of half-cell potentials for various materials [1]

Managing Dry Electrodes

One of the most frequent customer questions that comes up when debugging biopotential signal chains usually comes back to difficulty making good contact with dry electrodes. Sometimes this is just that one person in the office that tested it and can't seem to get a signal out. Dry electrodes look more like a polarizable electrode and may take some time to settle while very small amounts of perspiration start to build up. An easy check when having trouble getting a signal out is to start by wetting the electrodes and/or skin (water or hand sanitizer works just fine). If the signal shows up, then you know you are dealing with a contact issue.

A number of factors can lead to contact issues and should be considered as part of the design:

- Location of the electrodes, including any hair or dry skin (varies person to person)
- Electrode material and surface area making contact with skin (big difference between exercise electrodes and a smaller form factor like the [ADI VSM watch](#) shown in Figure 4)
- Pressure applied (like the tightness of a watch strap on the wrist)
- Leakage paths on the traces touching the electrodes (flux on the pcb)
- Input impedance of the measurement circuit as well as any other circuits touching the electrodes such as lead off detection (for example the addition of a pull-up/pull-down resistor will change the input impedance)

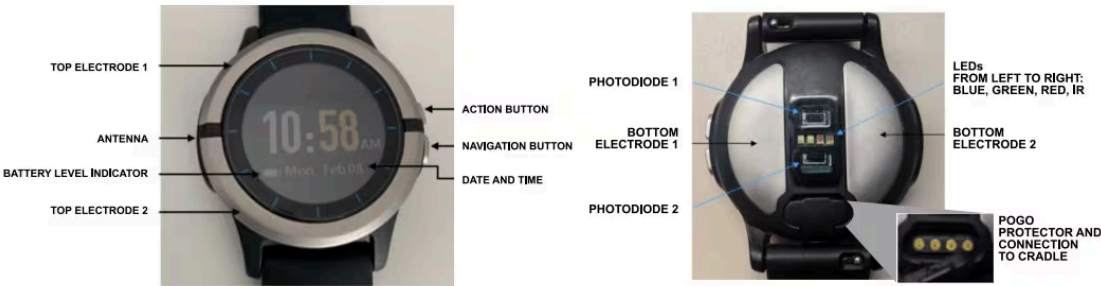


Figure 4 – Analog Devices Vital Signs Monitoring (VSM) Watch with top and bottom electrodes

Stay tuned for the next month’s blog where I will discuss electrode/lead off detection methods.

[1] Neuman, M. R. "Biopotential Electrodes." The Biomedical Engineering Handbook: Second Edition. Ed. Joseph D. Bronzino Boca Raton: CRC Press LLC, 2000.

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